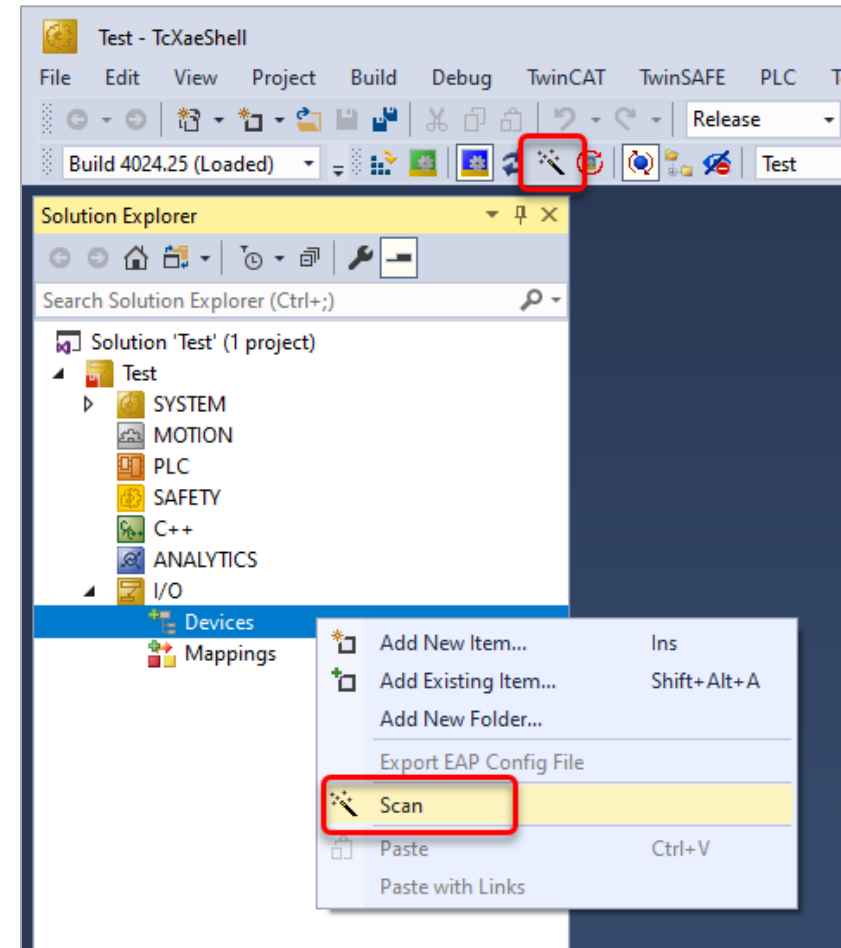
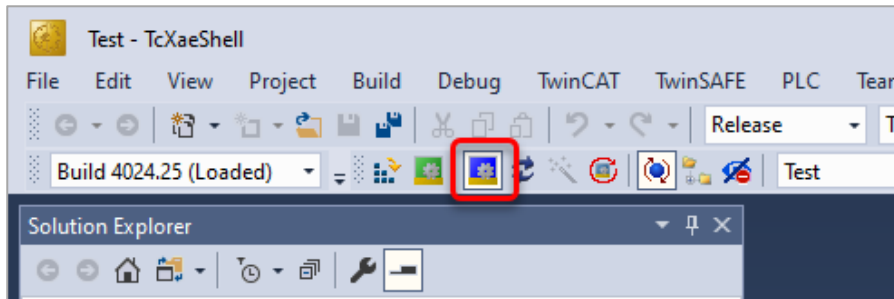


TwinCAT with a regular PD exchange axis

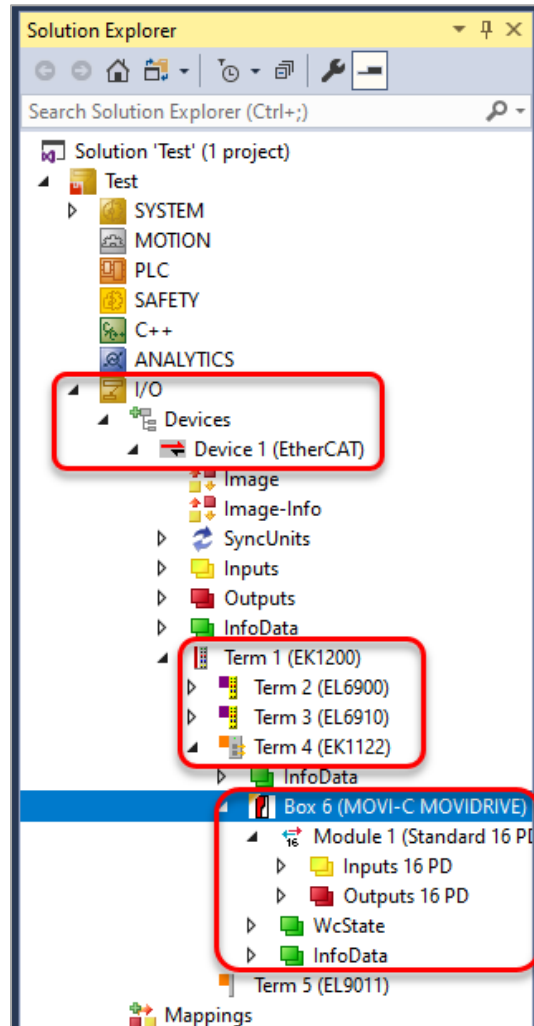
Adding an SBUSplus axis to TwinCAT

Hint: Not every step is shown!

Make sure to be in Config Mode to scan for devices

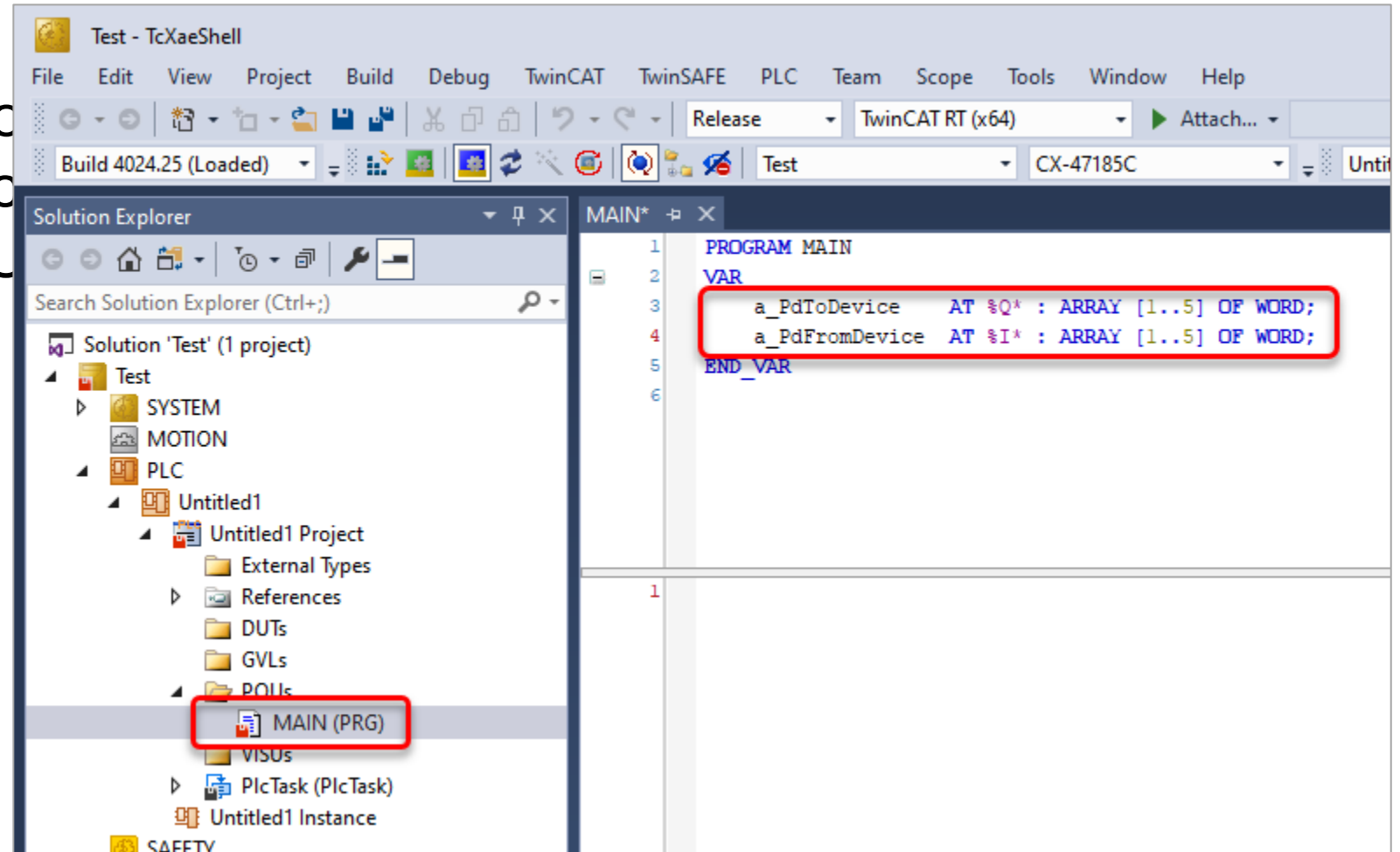


Example of a scanned device

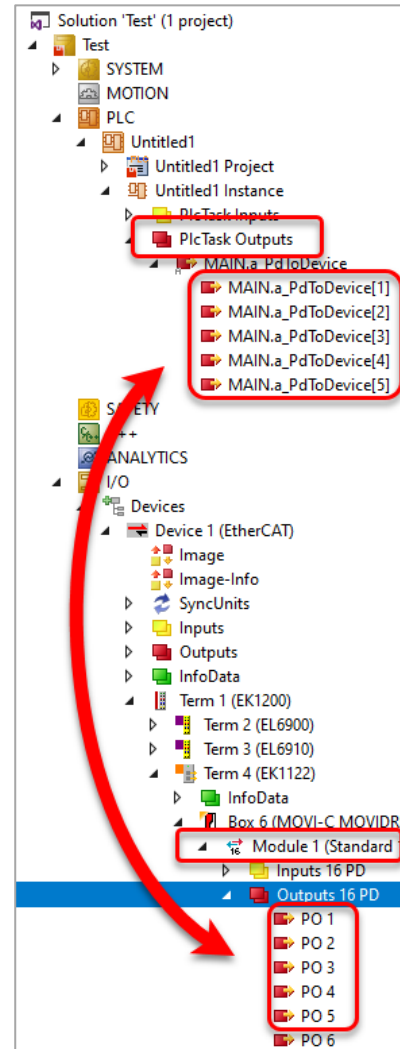
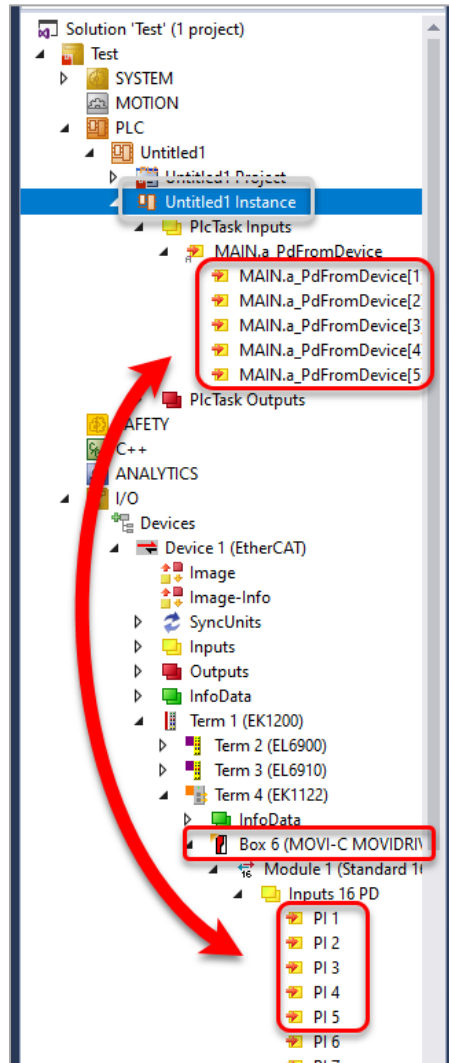


Create a PLC project and add variable instances

- Add a PLC project to your TwinCAT project and add e.g. to your POU mentioned variable declarations.

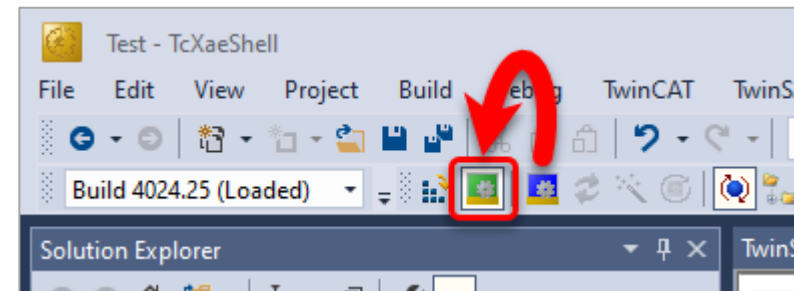
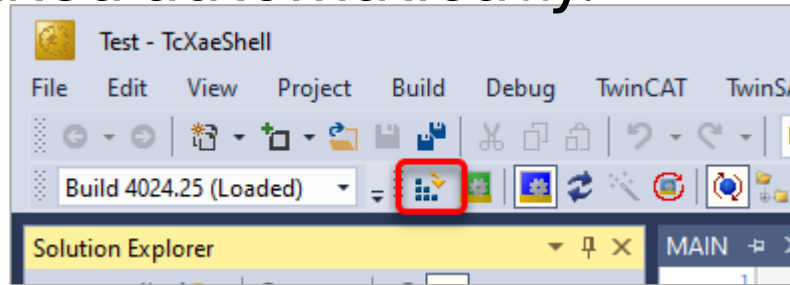


After building project, map instances to I/O device



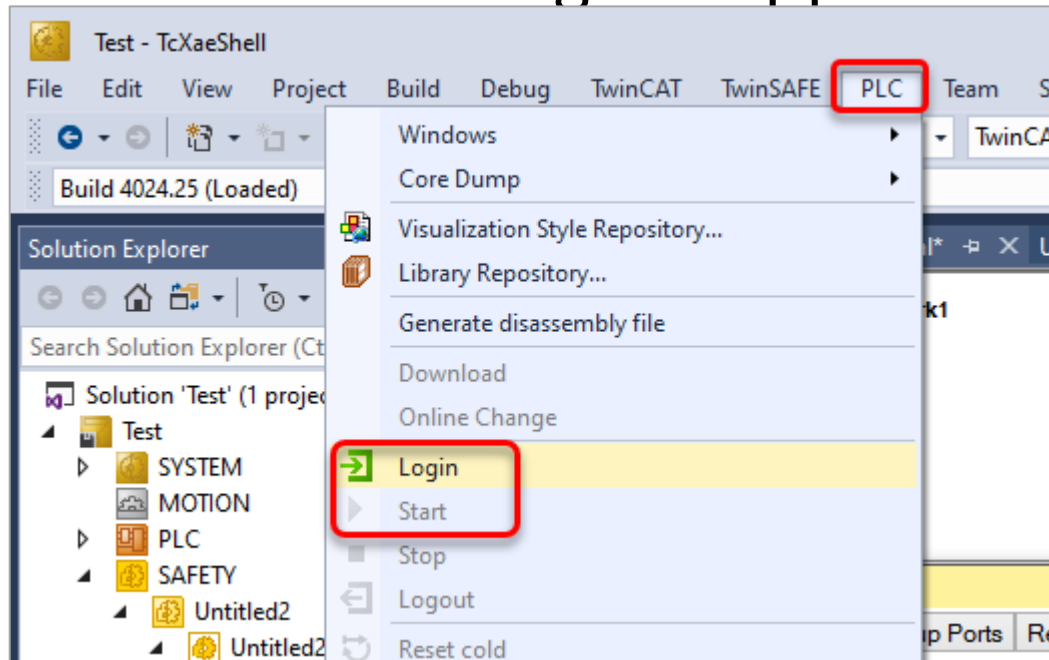
Activate/ download the configuration

- After download of config is completed, the run mode will be activated automatically.



Download and start the program

- Click “Login”
- After downloading the application, click “Start”



Check communication between device and PLC

e.g. on loaded MOVIKIT Velocity:

PC control

Activate...

User-Timeout

Off

Timeout-Counter

0

PD Aktualisierung

On

Process data overview

PLC

PLC output data

>>>>

PO 1: Control word

PO 2: Setpoint speed

PO 3: Acceleration

PO 4: Deceleration

PO 5: ... DO 03 ... DO 00

0x0003 hex

1111

1000

2000

0000 0001

mm/s

mm/s²

mm/s²

PLC input data

<<<<

0x3002 hex

0 mm/s

1

0.0 % nominal motor torque

0000 0000 0000 0000

mm/s

Drive

PI 1: Status word

PI 2: Actual speed

PI 3: Status/main fault - subfault

PI 4: Actual torque

PI 5: ... DI 13 .. DI 10, DI 07 ... DI 00

Expression	Type	Value
a_PdToDevice	ARRAY [1..5] OF W...	
a_PdToDevice[1]	WORD	3
a_PdToDevice[2]	WORD	1111
a_PdToDevice[3]	WORD	1000
a_PdToDevice[4]	WORD	2000
a_PdToDevice[5]	WORD	1
a_PdFromDevice	ARRAY [1..5] OF W...	
a_PdFromDevice[1]	WORD	12290
a_PdFromDevice[2]	WORD	0
a_PdFromDevice[3]	WORD	1
a_PdFromDevice[4]	WORD	0
a_PdFromDevice[5]	WORD	0

How-to add an SEW safety device to TwinSAFE

Add control/ status variables to MAIN.PRG

- These variables will be used later to control the safety function and see the actual status
- Add to MAIN.PRG's variable declaration
 - `xRunSafetyProg AT %Q* : BOOL;`
 - `xResetSafetyProg AT %Q* : BOOL;`
 - `xNoSTO_out AT %Q* : BOOL;`
 - `xSafetyReset_out AT %Q* : BOOL;`
 - `xSTOactive_in AT %I* : BOOL;`
 - `xSafetyError_in AT %I* : BOOL;`

Check and modify correct FSoE component

The screenshot shows the TwinCAT configuration environment. On the left, the Solution Explorer displays a project 'Test' with a device 'Device 1 (EtherCAT)' containing a 'Box 6 (MOVI-C MOVIDRIVE)'. The HW Config window on the right shows a table of modules. A red box highlights 'DriveSafety Axis 1' with module 'FSoE Safety IO 48 bit' and ID '0x00000180'. A red arrow points from this entry to the 'Module' list on the right, which also has a red box around 'FSoE Safety IO 48 bit' with ID '0x00000180'.

Slot	Module	ModuleIdent
Axis 1	Standard 16 PD	0x00000100
DriveSafety Axis 1	FSoE Safety IO 48 bit	0x00000180
Axis 2		
DriveSafety Axis 2		

Module	ModuleIdent	Description
FSoE Safety IO 48 bit	0x00000180	FSoE Safety IO 48 bit
FSoE Safety IO 48 bit + INT + DINT	0x00000181	FSoE Safety IO 48 bit + INT + DINT
Failsafe Container 10 Byte	0x000001F0	Failsafe 10 Byte
Failsafe Container 14 Byte	0x000001F1	Failsafe 14 Byte
Failsafe Container 28 Byte	0x000001F2	Failsafe 28 Byte
Failsafe Container 18 Byte	0x000001F3	Failsafe 18 Byte
Failsafe Container 22 Byte	0x000001F4	Failsafe 22 Byte
Failsafe Container 42 Byte	0x000001F5	Failsafe 42 Byte

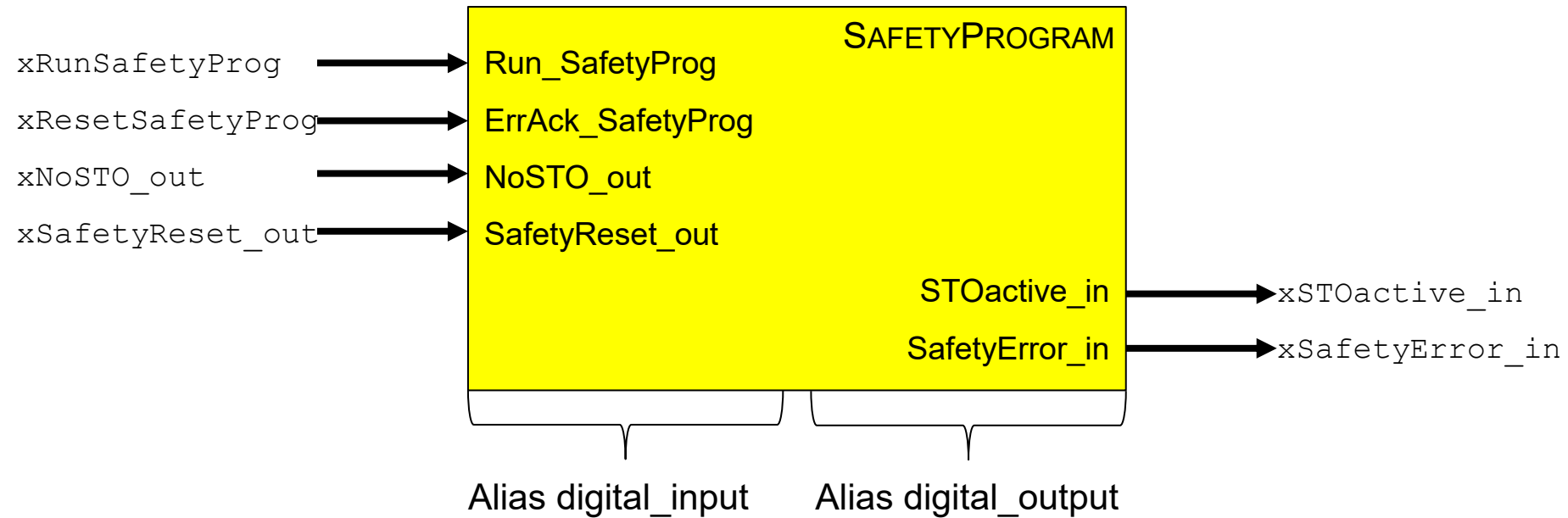
Name	Online	Type	Size	> Addr...	In/Out	User ID	Linked to
PI 1		UINT	2.0	75.0	Input	0	
PI 2		UINT	2.0	77.0	Input	0	
PI 3		UINT	2.0	79.0	Input	0	
PI 4		UINT	2.0	81.0	Input	0	
PI 5		UINT	2.0	83.0	Input	0	
PI 6		UINT	2.0	85.0	Input	0	

- 0x00000180 -> SBUSplus: Safety cards with regular safety profile “Technology”
- 0x00000181 -> SBUSplus: Safety cards with extended safety profile “System”
- 0x04020081 -> CiA402: Safety cards with regular safety profile “Technology”
- 0x04020082 -> CiA402 : Safety cards with extended safety profile “System”

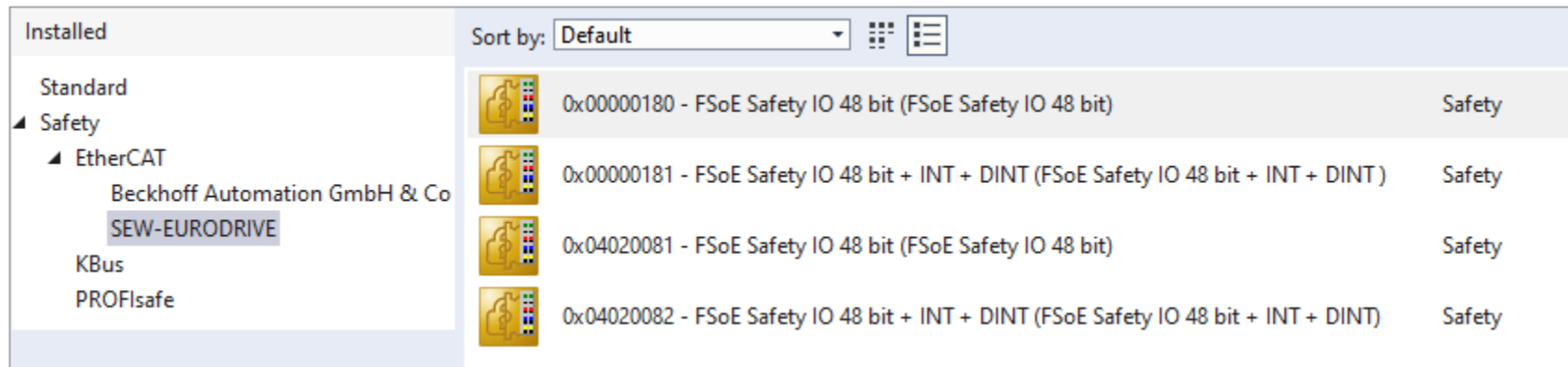
What is an “Alias”?

- It's a placeholder to link variables and safety devices between the unsafe world and the safety program
- Unsafe output variables are inputs in the safety program
- Safety program output variables are unsafe input variables

Creating Aliases for variables

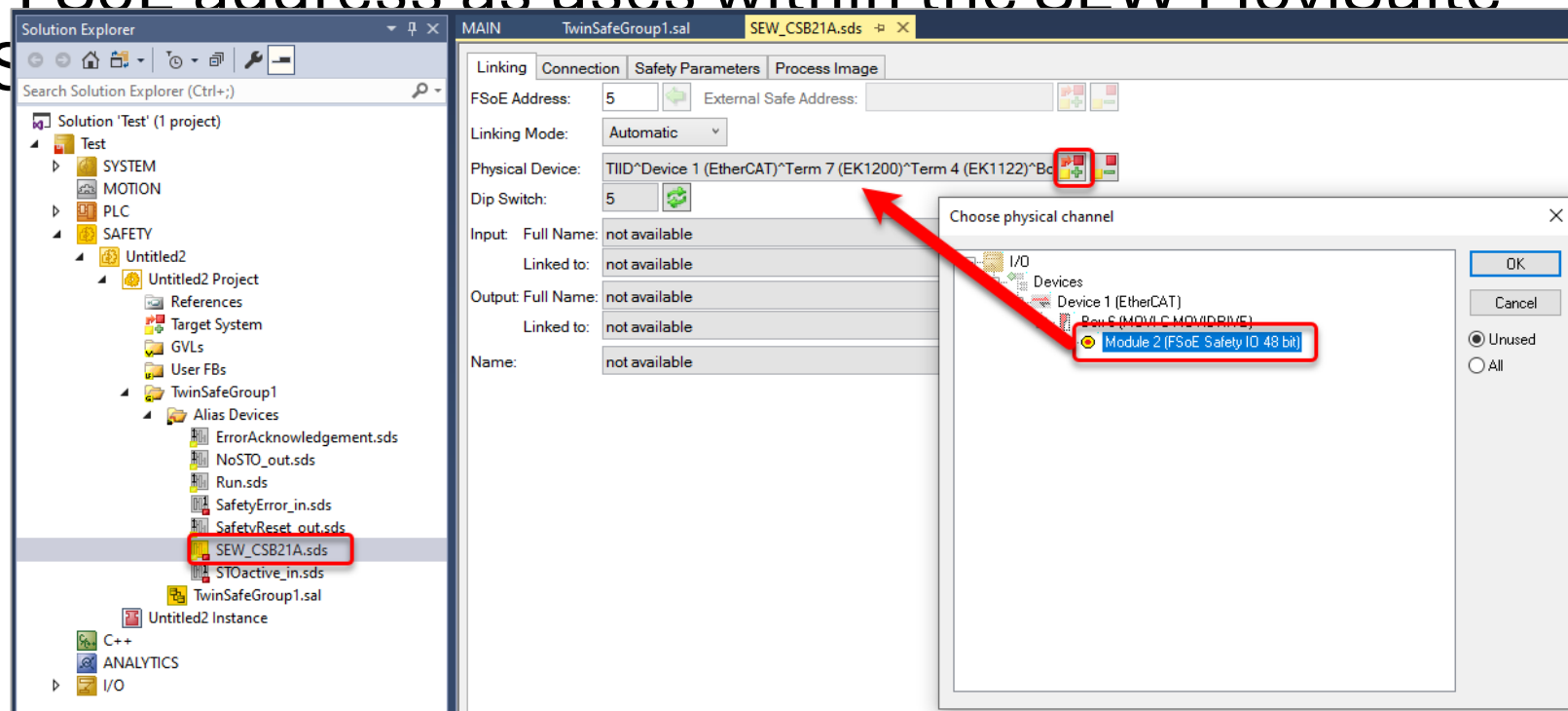


Create an Alias with correct safety device

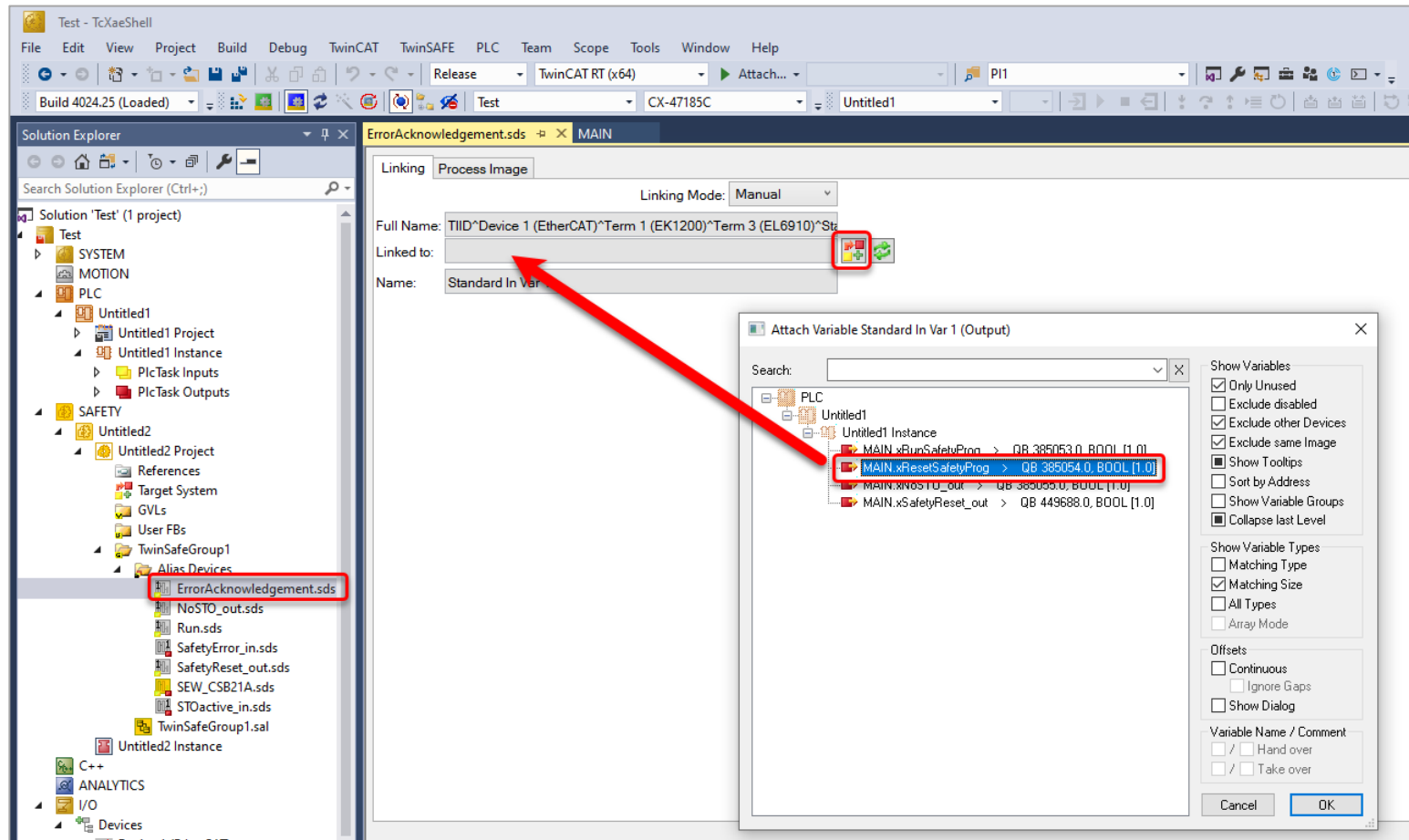


Link safety device

- Link safety device from I/O to Alias
- Set same FSoE address as uses within the SEW MoviSuite Assist..CS



Link all unsafe variables to safety program variables



Rename safety bits of process image of safety device

- For better understanding, rename bits. Watch for correct bit position!

The screenshot displays the TwinCAT software interface for configuring a safety device. The central panel shows the 'Process Image' configuration for 'TwinSafeGroup1.sal'. The 'Inputs' table lists 21 bits, with 'CSB_STOActive_in' at position 0.0 and 'CSB_SafetyError_in' at position 0.7 highlighted by red boxes. The 'Outputs' table lists 21 bits, with 'CSB_STO_out' at position 0.0 and 'CSB_SafetyReset_out' at position 0.7 highlighted by red boxes. The Solution Explorer on the left shows the project structure, with 'SEW_CSB21A.sds' highlighted under the 'Alias Devices' folder.

Name	Type	Size	Position	Map Info Data
CSB_STOActive_in	BIT	0.1	0.0	
Input_02	BIT	0.1	0.1	
Input_03	BIT	0.1	0.2	
Input_04	BIT	0.1	0.3	
Input_05	BIT	0.1	0.4	
Input_06	BIT	0.1	0.5	
Input_07	BIT	0.1	0.6	
CSB_SafetyError_in	BIT	0.1	0.7	
Input_09	BIT	0.1	1.0	
Input_10	BIT	0.1	1.1	
Input_11	BIT	0.1	1.2	
Input_12	BIT	0.1	1.3	
Input_13	BIT	0.1	1.4	
Input_14	BIT	0.1	1.5	
Input_15	BIT	0.1	1.6	
Input_16	BIT	0.1	1.7	
Input_17	BIT	0.1	2.0	
Input_18	BIT	0.1	2.1	
Input_19	BIT	0.1	2.2	
Input_20	BIT	0.1	2.3	
Input_21	BIT	0.1	2.4	

Name	Type	Size	Position	Map Info Data
CSB_STO_out	BIT	0.1	0.0	
Output_02	BIT	0.1	0.1	
Output_03	BIT	0.1	0.2	
Output_04	BIT	0.1	0.3	
Output_05	BIT	0.1	0.4	
Output_06	BIT	0.1	0.5	
Output_07	BIT	0.1	0.6	
CSB_SafetyReset_out	BIT	0.1	0.7	
Output_09	BIT	0.1	1.0	
Output_10	BIT	0.1	1.1	
Output_11	BIT	0.1	1.2	
Output_12	BIT	0.1	1.3	
Output_13	BIT	0.1	1.4	
Output_14	BIT	0.1	1.5	
Output_15	BIT	0.1	1.6	
Output_16	BIT	0.1	1.7	
Output_17	BIT	0.1	2.0	
Output_18	BIT	0.1	2.1	
Output_19	BIT	0.1	2.2	
Output_20	BIT	0.1	2.3	
Output_21	BIT	0.1	2.4	

Create in “Variable Mapping” links

- Aliases have to be linked to an internal name which can be used

The screenshot displays the TwinCAT software interface. On the left, the Solution Explorer shows a project structure with 'TwinSafeGroup1.sal' selected. The main window shows the 'Variable Mapping' for 'Network1'. The table below lists the mappings:

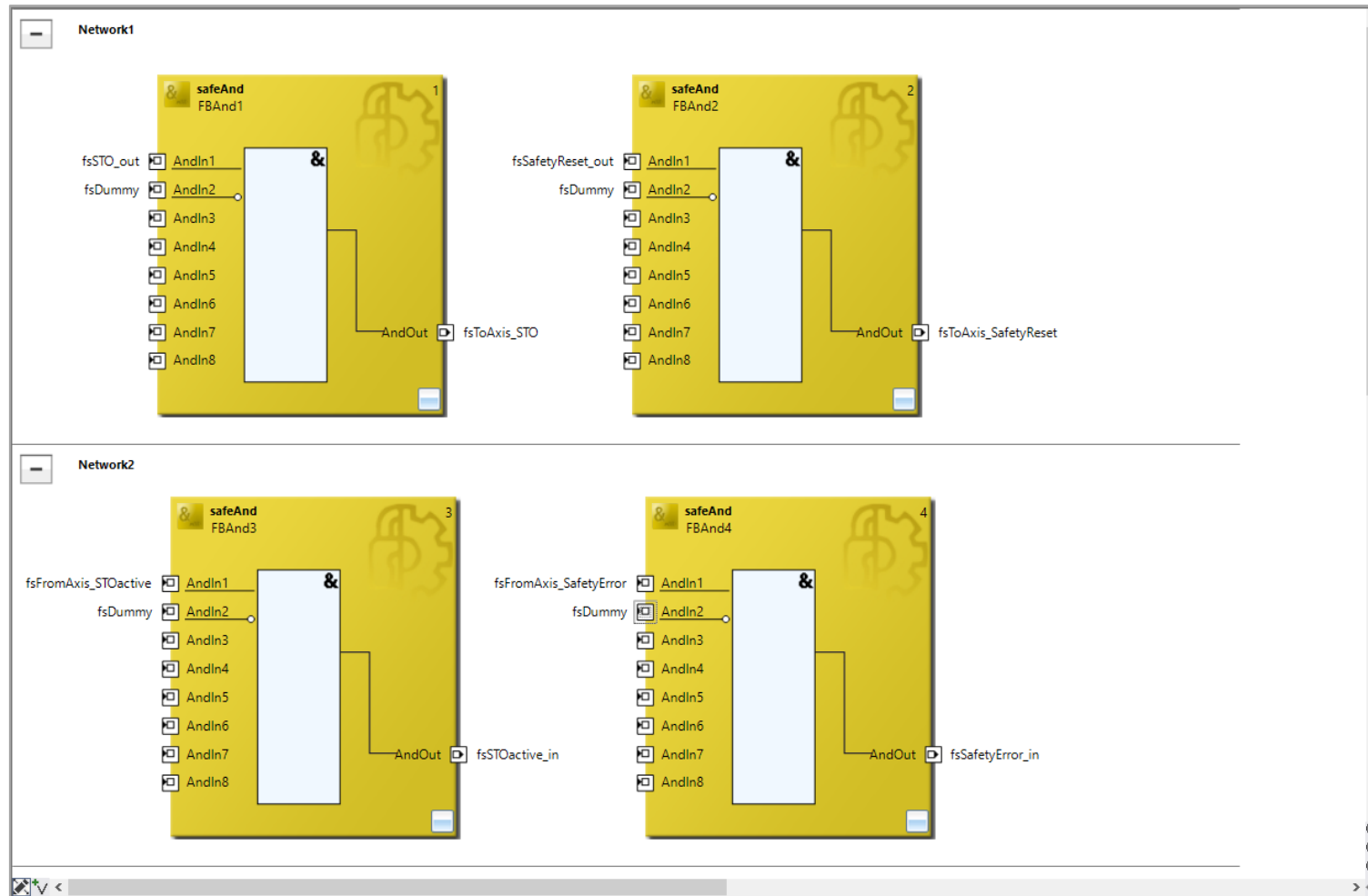
Variable	Scope	Assignment	Usages	Online Value	Comment
GroupPort_ErrAck	Local	ErrorAcknowledgement.In (TwinSafeGroup1)	TwinSafeGroup1.Err Ack		
GroupPort_RunStop	Local	Run.In (TwinSafeGroup1)	TwinSafeGroup1.Run/Stop		
fsSTO_out	Local	NoSTO_out.In (TwinSafeGroup1)			
fsSafetyReset_out	Local	SafetyReset_out.In (TwinSafeGroup1)			
fsSTOActive_in	Local		STOActive_in.Out (TwinSafeGroup1)		
fsSafetyError_in	Local		SafetyError_in.Out (TwinSafeGroup1)		

Red annotations highlight specific elements:

- A red circle around 'fsSTO_out' with a callout 'Give a name'.
- A red box around 'NoSTO_out.In (TwinSafeGroup1)' with a callout 'Alias - Digital Inputs'.
- A red box around 'STOActive_in.Out (TwinSafeGroup1)' with a callout 'Alias - Digital Outputs'.

The bottom of the window shows tabs for 'Safety Project Online View', 'Variable Mapping', 'Error List', and 'Output'.

Example of safety program



Full linked “Variable Mapping” with example safety program

Safety Project Online View					
Variable Mapping					
Error List					
Output					
Variables					
Group Ports					
Replacement Values					
Max Start Deviation					
Group					
Variable	Scope	Assignment	Usages	Online	
GroupPort_RunStop	Local	Run.In (TwinSafeGroup1)	TwinSafeGroup1.Run/Stop		
GroupPort_ErrAck	Local	ErrorAcknowledgement.In (TwinSafeGroup1)	TwinSafeGroup1.Err Ack		
fsToAxis_STO	Local	TwinSafeGroup1.Network1.FBAnd1.AndOut	SEW_CSB21A.CSB_STO_out (TwinSafeGroup1)		
fsToAxis_SafetyReset	Local	TwinSafeGroup1.Network1.FBAnd2.AndOut	SEW_CSB21A.CSB_SafetyReset_out (TwinSafeGroup1)		
fsSTOactive_in	Local	TwinSafeGroup1.Network2.FBAnd3.AndOut	STOactive_in.Out (TwinSafeGroup1)		
fsSTO_out	Local	NoSTO_out.In (TwinSafeGroup1)	TwinSafeGroup1.Network1.FBAnd1.AndIn1		
fsSafetyReset_out	Local	SafetyReset_out.In (TwinSafeGroup1)	TwinSafeGroup1.Network1.FBAnd2.AndIn1		
fsSafetyError_in	Local	TwinSafeGroup1.Network2.FBAnd4.AndOut	SafetyError_in.Out (TwinSafeGroup1)		
fsFromAxis_STOactive	Local	SEW_CSB21A.CSB_STOactive_in (TwinSafeGroup1)	TwinSafeGroup1.Network2.FBAnd3.AndIn1		
fsFromAxis_SafetyError	Local	SEW_CSB21A.CSB_SafetyError_in (TwinSafeGroup1)	TwinSafeGroup1.Network2.FBAnd4.AndIn1		
fsDummy	Local	SEW_CSB21A.CSB_Dummy (TwinSafeGroup1)	TwinSafeGroup1.Network1.FBAnd1.AndIn2 TwinSafeGroup1.Network1.FBAnd2.AndIn2 TwinSafeGroup1.Network2.FBAnd3.AndIn2 TwinSafeGroup1.Network2.FBAnd4.AndIn2		

Check out the SEW safety device checksum (“iPar_CRC”)

Device properties

Drive trains

- Drive train DT2
- Optimization DT2

Functions

- Inputs/outputs
- Setpoints
- Actual values
- Drive functions
- Technology functions
- Monitoring functions
- MOVISAFE® CS..

Diagnostics

- Status
- Process values
- Fault memory
- Communication
- MOVIKIT® diagnostics
- MOVISAFE® CS..**

MOVIKIT® Velocity Drive

6.4.46 F-communication

Configuration

Function	Value
Protocol type	FSOE
F-address	5
ParCRCBus (iPar_CRC)	0xD9FE E686

Status

Diagnostics information

Safe data exchange

Reset diagnostic parameters

Telegrams

Telegram data

	Telegram data
Rx	0x36 0x01 0x00 0x4A 0x97 0x00 0x00 0xEF 0x94 0x00 0x00 0xCA 0x82 0x08 0x00
Tx	0x36 0x08 0x00 0x4C 0x81 0x00 0x00 0x07 0x92 0x00 0x00 0x22 0x84 0x08 0x00

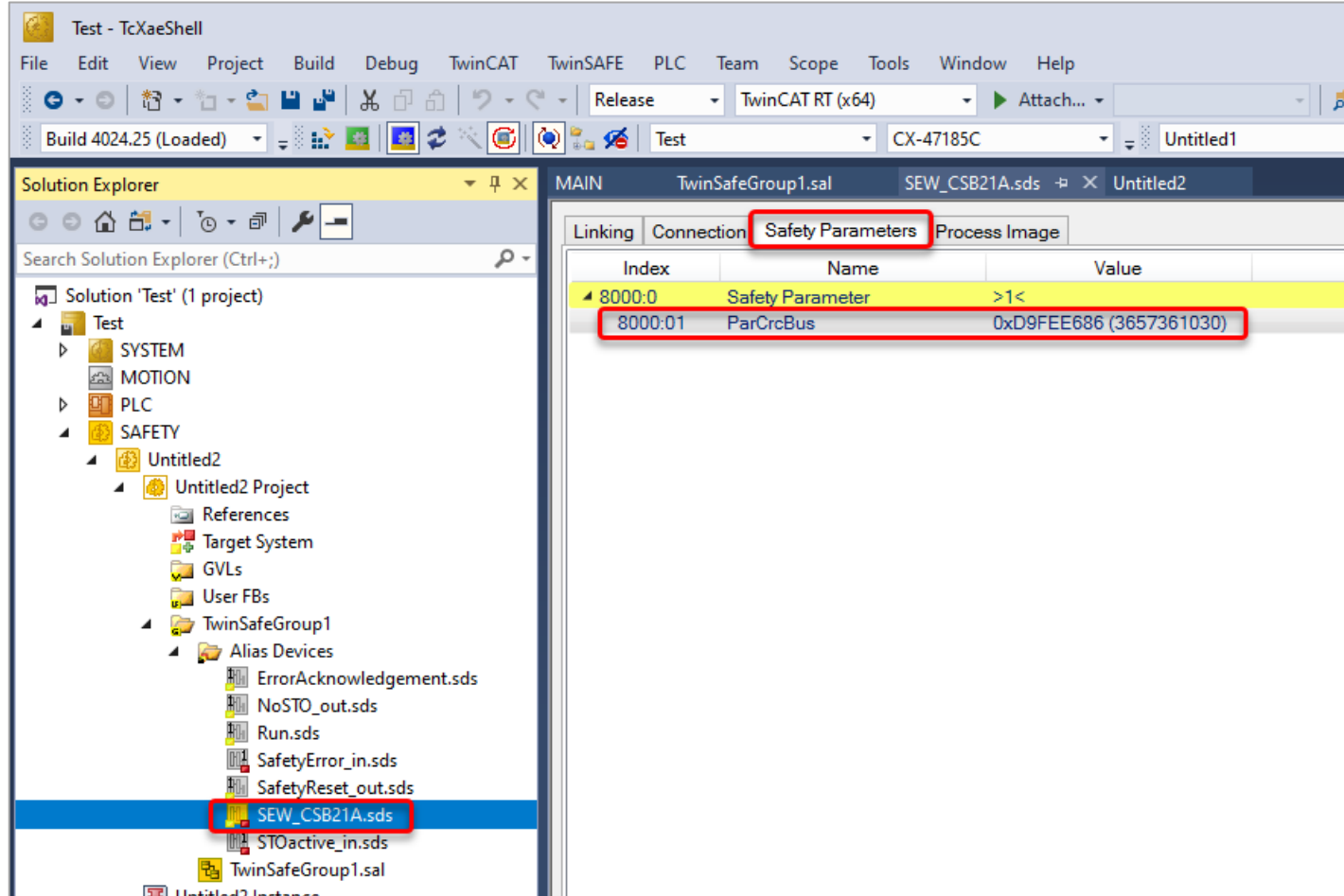
Telegram counter

Function	Value
Valid received	1307586
Repeated received	17825460
Corrupt received	0
Sent	1307588

Data exchange time response

Function	Value
Timeout	100 ms
Maximum	60 ms
Average	59 ms
Minimum	48 ms

Add checksum to safety device



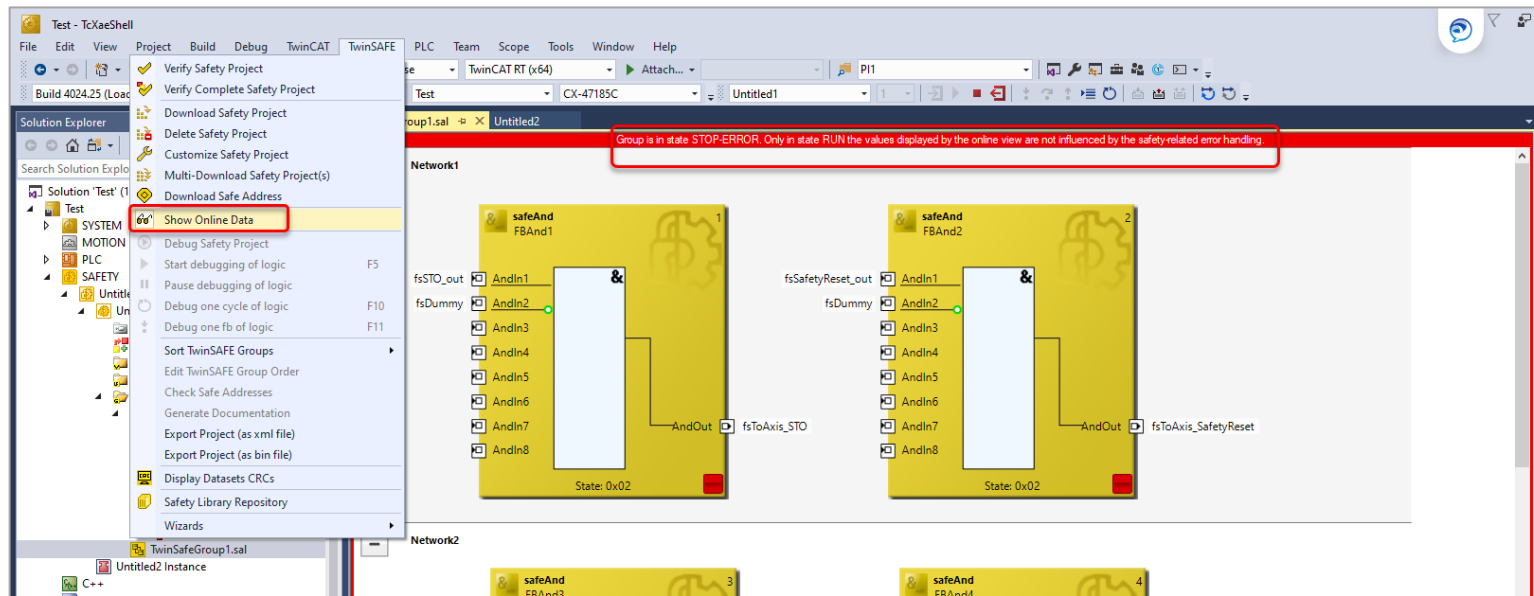
Download of safety program

- Make sure that safety program is selected that download option is available
- When download is clicked, the safety program will be verified automatically



Check status of safety program

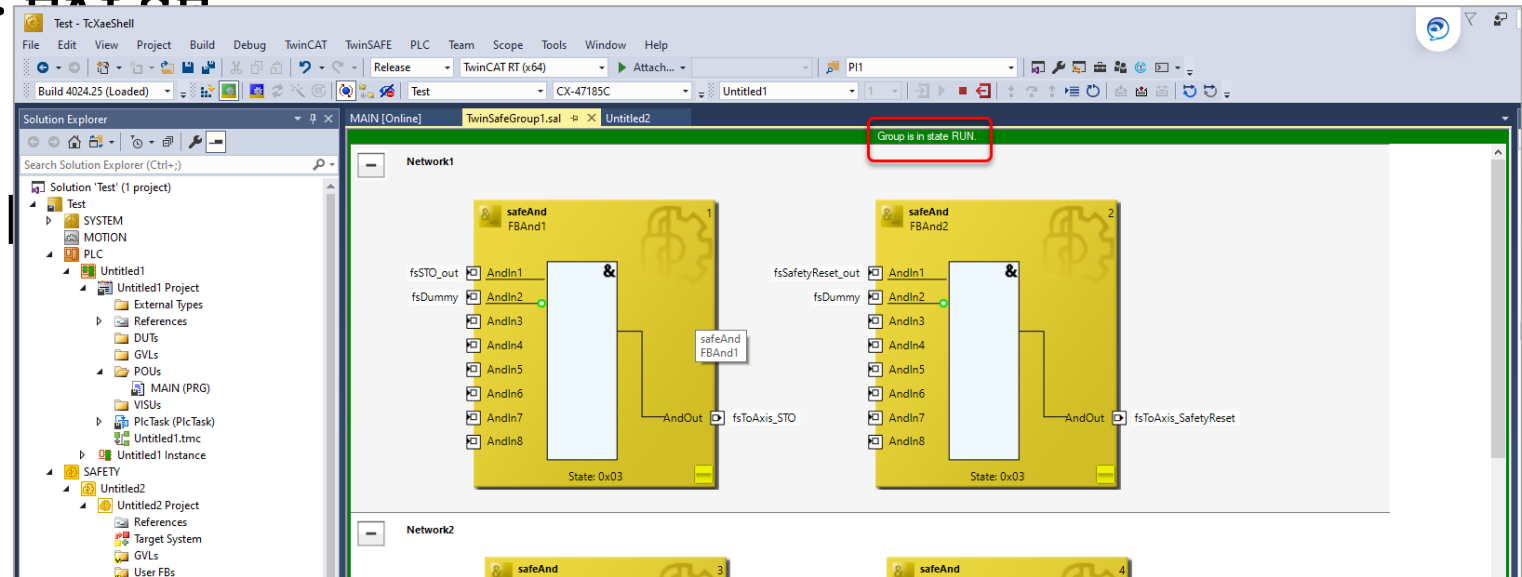
- Click “TwinSAFE”, then “Show Online Data”
- Probably program is in STOP



Reset and run safety program

- Set in MAIN.PRG variables:
 - xResetSafetyProg -> **TRUE**
 - xRunSafetyProg -> **TRUE**
 - xResetSafetyProg -> **FALSE**

- Safety program should show now a green frame



Toggle bits in MAIN.PRG

- Toggle “xNoSTO_out” and see the result in

MAIN [Online] X TwinSafeGroup1.sal Untitled2				
Test.Untitled1.MAIN				
Expression	Type	Value	Prepared value	Addr
⊕ a_PdToDevice	ARRAY [1..5] OF W...			%Q*
⊕ a_PdFromDevice	ARRAY [1..5] OF W...			%I*
⊕ xRunSafetyProg	BOOL	TRUE		%Q*
⊕ xResetSafetyProg	BOOL	FALSE		%Q*
⊕ xNoSTO_out	BOOL	TRUE		%Q*
⊕ xSafetyReset_out	BOOL	FALSE		%Q*
⊕ xSTOactive_in	BOOL	FALSE		%I*
⊕ xSafetyError_in	BOOL	FALSE		%I*

				Value	Prepared value	Addr
						%Q*
						%I*
⊕ xRunSafetyProg	BOOL	TRUE				%Q*
⊕ xResetSafetyProg	BOOL	FALSE				%Q*
⊕ xNoSTO_out	BOOL	FALSE				%Q*
⊕ xSafetyReset_out	BOOL	FALSE				%Q*
⊕ xSTOactive_in	BOOL	TRUE				%I*
⊕ xSafetyError_in	BOOL	FALSE				%I*